

Test 37: (1/2)! via ? ? half-factorial & GUE normalization

Verdict: PASS | Generated: 2026-09-02 23:33:21 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT: $\Gamma(1/2) = \sqrt{\pi}$, $(1/2)! = \Gamma(3/2) = \sqrt{\pi}/2$, ladder + recurrence (suite's own Lanczos lgamma); UNIQUENESS of the Γ -construction via the reflection self-dual fixed point $\Gamma(1/2)\Gamma = \pi$

METHOD: EXACT: $\Gamma(1/2) = \sqrt{\pi}$, $(1/2)! = \Gamma(3/2) = \sqrt{\pi}/2$, ladder + recurrence (suite's own Lanczos lgamma); UNIQUENESS of the Γ -construction via the reflection self-dual fixed point $\Gamma(1/2)\Gamma = \pi$

Test 37 HARDCORE: 256-bit DE $\Gamma(1/2)$ err 1.20e-28, Float64 round-off $\Gamma(1/2)$ 5.83e-16 / $(1/2)!$ 2.94e-16, N=32/?? err 2.51e-26, ladder/self-dual/recurrence exact.

Result [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- `plots/plot\_01` (.png 600dpi / .svg / .pdf)

Plots

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 37 --no-two-pass`

Independent verification guide: `indep_comp_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(no log_comp entries)

FULL COMPUTATION LOG

??

CONSOLE CAPTURE

TEST 37 (pass 2, HARDCORE): (1/2)! via ? ? 256-bit precision audit

??

DE reference: $|\Gamma(1/2)_{256} - \sqrt{\pi}| = 1.20\text{e-}28$ (independent quadrature)

$\Gamma(1/2)$: Float64 vs 256-bit DE rel.err = 5.83e-16

$(1/2)!$: Float64 vs 256-bit DE rel.err = 2.94e-16

N = $1/[\frac{1}{2} \cdot \Gamma(3/2) \cdot (1/2)!]$ at 256-bit vs 32/?: rel.err = 2.51e-26

ladder $\Gamma(5/2)_{DE} = 3\sqrt{\pi}/4$: rel.err = 4.78e-25

$\Gamma(1/2)_{DE} = \pi$ (reflection self-duality): rel.err = 1.35e-28

recurrence $(1/2)!_{DE} = \frac{1}{2} \cdot \Gamma(1/2)_{DE}$: rel.err = 2.51e-26

??

Float64 round-off PASS ? $|\Gamma(1/2)|=5.83\text{e-}16$, $|(1/2)!|=2.94\text{e-}16$ relative (machine-? scale)

32/?? identity (256-bit) PASS ? rel.err 2.51e-26 ? exact at 256-bit

ladder + self-duality + recurrence (256-bit) PASS ? worst rel.err 4.78e-25 across the three identities

Test 37b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS